07 — Cognitive Assessment: Full Prep

The cognitive test measures reasoning **speed and accuracy**, not programming. Four sections: **Numerical, Verbal, Abstract, and General Mental Ability.** Most people don't finish — so **accuracy on what you attempt beats rushing.** 30–60 seconds per question, guess-and-move-on if stuck (guessing beats blank when there's no penalty — confirm on the day).

This guide teaches each section with **worked examples**, then gives you **practice questions with answers** at the end. Do the practice under a timer.

SECTION 1 — NUMERICAL REASONING

1a. Number sequences — find the rule

Check these rules **in this order** (first that fits wins):

Rule	Looks like	Example
1. Constant difference (+d or −d)	steady steps	2, 5, 8, 11 → +3 → **14**
2. Growing difference	steps get bigger	1, 2, 4, 7, 11 → +1,+2,+3,+4 → **16**
3. Multiply (×k)	numbers explode	3, 6, 12, 24 → ×2 → **48**
4. Alternating (two patterns)	zig-zag	1, 10, 2, 20, 3 → odds +1, evens ×2 → **30**
5. Squares / cubes / primes	familiar numbers	1, 4, 9, 16 → squares → **25**
6. Add the two before (Fibonacci)	each = sum of prev two	1, 1, 2, 3, 5, 8 → **13**

Worked example: `4, 9, 16, 25, ?`

- Try constant difference: +5, +7, +9 → not constant, but the differences grow by 2 → could be rule 2.
- Try squares: 2^2=4, 3^2=9, 4^2=16, 5^2=25 → yes! Next is 6^2 = **36**.

Worked example: `2, 6, 18, 54, ?`

- Differences +4, +12, +36 — not constant. Numbers grow fast → try ×: 2×3=6, 6×3=18, 54×3 = **162**.

1b. Percentages — the fast tricks

Build any percentage from these three:

- **10%** = divide by 10 (move the decimal one left). 10% of 250 = 25
- **5%** = half of 10%. 5% of 250 = 12.5
- **1%** = divide by 100. 1% of 250 = 2.5

So **15% of 250** = 10% + 5% = 25 + 12.5 = **37.5**

The swap trick — memorize this: `x% of y = y% of x`

- 8% of 25 → flip → 25% of 8 = **2** (much easier!)
- 4% of 50 → 50% of 4 = **2**

Increase / decrease:

- Increase by 20% → ×1.20
- Decrease by 20% → ×0.80
- "What % is 30 of 120?" → 30/120 = 0.25 = **25%**

1c. Ratios

Total = sum of the ratio parts. Split the total into that many equal parts.

Worked example: share 40 sweets in ratio 3:2.

- Parts: 3 + 2 = 5
- One part = 40 ÷ 5 = 8
- So 3×8 = **24** and 2×8 = **16** (check: 24+16 = 40 ✓)

1d. Averages (mean)

Average = total ÷ how many.

- Average of 4, 8, 6 → (4+8+6) ÷ 3 = 18 ÷ 3 = **6**
- Reverse: "average of 3 numbers is 10, two are 8 and 9, find the third" → total must be 30; 30 − 8 − 9 = **13**

1e. Reading tables / charts

- Read the **question first**, then find only the numbers you need.
- Watch **units** (thousands? percentages? months?).
- "How much MORE did A make than B?" → subtract. "How many TIMES?" → divide.

SECTION 2 — VERBAL REASONING (True / False / Cannot Say)

You get a short passage, then statements. Decide:

- **True** — the passage says it, or it clearly must follow from what's said.
- **False** — the passage says the opposite.
- **Cannot Say** — the passage doesn't give enough to decide.

The one rule that matters

Use ONLY the passage. Your own knowledge is a trap. If it isn't stated or directly implied, the answer is **Cannot Say** — even if you know it's true in real life.

Worked example

> *Passage:* "The library opens at 9 am on weekdays. On Saturdays it opens at 10 am. It is closed on Sundays. Members may borrow up to 5 books at a time."

Statement	Answer	Why
"The library opens at 9 am on Mondays."	**True**	Monday is a weekday → stated
"The library is open on Sundays."	**False**	passage says closed
"Non-members can borrow books."	**Cannot Say**	passage only mentions members — nothing about non-members
"The library has more than 1000 books."	**Cannot Say**	not mentioned at all (even if likely true!)
"On Saturday it opens later than on Tuesday."	**True**	10 am vs 9 am → follows from stated facts

The trap in that example: "more than 1000 books" *feels* true — libraries are big — but the passage never says it. → Cannot Say.

Tips

- **"All", "always", "never", "only"** are strong words. One counter-example makes them False.
- If a statement adds NEW information not in the passage → almost always **Cannot Say**.
- Don't argue with the passage — treat it as the only truth that exists.

SECTION 3 — ABSTRACT / LOGICAL REASONING (shapes & patterns)

"Which shape comes next?" / "Which is the odd one out?" / "Complete the grid." No numbers or words — just visual patterns. You look for **what is changing** from one shape to the next.

What to check (run through these)

1. **Rotation** — is the shape turning? (90° each step? clockwise?)
2. **Count** — are dots/sides/lines increasing or decreasing? (1, 2, 3, …)
3. **Shading / colour** — flipping black↔white? filling in?
4. **Position** — is something moving around the box (corner to corner)?
5. **Size** — growing or shrinking?
6. **Adding / removing** — does a new element appear each step?

Often **two things change at once** (e.g. rotating AND gaining a dot). Track each separately.

Worked example (described)

> A square with an arrow. Step 1: arrow points up. Step 2: points right. Step 3: points down. What's step 4?

The arrow rotates 90° clockwise each step: up → right → down → **left**. Answer: arrow left.

Odd-one-out

Find the rule that **four of five share**, and the one that breaks it.

> Four shapes have an even number of sides (square, hexagon…), one has 5 (pentagon) → the pentagon is the odd one out.

Check: number of sides, shading, symmetry, number of shapes inside, orientation.

Matrix (3×3 grid, one cell missing)

Read it like a table: look **across each row** for the rule, then **down each column**. The missing cell must satisfy both.

SECTION 4 — GENERAL MENTAL ABILITY / SPEED

Mixed quick questions and **string/number matching** (comparing two things for an exact match).

Matching — the accuracy drill

> Are these identical? `48273901` vs `48279301` Compare in **chunks**: `4827` = `4827` ✓, then `3901` vs `9301` ✗ → **different**.

Tip: compare left-to-right in small groups. Most errors are transposed digits (`39` vs `93`).

Quick logic

> "All roses are flowers. Some flowers fade quickly. Do all roses fade quickly?" **No / Cannot say** — "some flowers" isn't "all flowers", so you can't conclude roses do. (Same "use only what's stated" rule as verbal.)

PRACTICE SET (do under a timer — ~45 seconds each)

Sequences — find the next number:

1. 3, 7, 11, 15, ?
2. 2, 4, 8, 16, ?
3. 1, 4, 9, 16, ?
4. 5, 10, 20, 40, ?
5. 1, 1, 2, 3, 5, ?
6. 100, 90, 81, 73, ? (differences: −10, −9, −8, …)

Percentages: 7. 20% of 150 = ? 8. 25% of 80 = ? 9. 6% of 50 = ? (use the swap trick) 10. What % is 15 of 60?

Ratios & averages: 11. Share 60 in ratio 4:1. What's the bigger share? 12. Average of 10, 20, 30, 40 = ? 13. Three numbers average 20. Two are 15 and 25. The third?

Verbal (passage: "All staff must wear a badge. Visitors sign in at reception. The office closes at 6 pm."): 14. "Staff wear badges." → ? 15. "Visitors must wear badges." → ? 16. "The office is open at 5 pm." → ? 17. "There are 50 staff." → ?

Logic: 18. "Some cats are black. All black things absorb heat. Do some cats absorb heat?" → ? 19. Odd one out: circle, square, triangle, sphere, pentagon → ? 20. Matching — identical? `PQ7R3T` vs `PQ7R8T` → ?

ANSWERS

1. **19** (+4) · 2. **32** (×2) · 3. **25** (squares, 5^2) · 4. **80** (×2) · 5. **8** (Fibonacci) · 6. **66** (−7 next: differences shrink by 1)
2. **30** (10%=15, ×2) · 8. **20** (quarter of 80) · 9. **3** (swap → 50% of 6) · 10. **25%** (15/60)
3. **48** (parts 4+1=5, one part 12, bigger = 4×12) · 12. **25** (100÷4) · 13. **20** (total 60 − 15 − 25)
4. **True** · 15. **Cannot Say** (only staff are stated to wear badges) · 16. **True** (closes at 6, so open at 5) · 17. **Cannot Say** (number not given)
5. **Yes** (some cats are black → those black cats absorb heat → some cats do) · 19. **sphere** (the only 3-D shape) · 20. **Different** (`R3T` vs `R8T`)

The night-before / morning checklist

- Do 2–3 timed practice sets in the days before (search "free numerical reasoning test", "free abstract reasoning test" — 123test.com, AssessmentDay, Practice Aptitude Tests).
- **Sleep well** — cognitive scores drop measurably when tired.
- **Speed rule:** 30–60 seconds per question. Stuck? Guess and move on (no penalty = never blank).
- **Accuracy over completion** — these are designed so most people don't finish.
- Keep scratch paper for the numerical section — don't do it all in your head.
- Verbal: **only the passage**. Logic: **only what's stated**. Ignore real-world knowledge.